

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Civil) (Credit System) Degree Examinations

May - June 2023

20CVE602 - SUSTAINABLE CONSTRUCTION MATERIALS AND METHODS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the properties and technical uses of i) translucent concrete	10	L*2	1	1
	ii) laminated timber.				
b)	Explain i) Aluminium forms ii) Fiberglass shuttering.	10	L2	2	1
2. a)	Explain i) self healing concrete ii) hydro ceramics.	10	L2	1	1
b)	Explain i) modular forms ii) pollution absorbing bricks.	10	L2	2	1
3. a)	Explain the suitability and benefits of modular bamboo as structural members.	10	L2	1	1
b)	Explain the suitability, working and advantages of tunnel forms.	10	L2	2	1
Unit – II					
4. a)	Summarise the techniques used in 3d volumetric construction and thin joint masonry technique.	10	L2	3	1
b)	Explain sustainable construction management techniques with examples.	10	L2	4	1
5. a)	Summarise the techniques used in precast flat panel-modules, pre-cast foundation technique.	10	L2	3	1
b)	Explain any two sustainable construction methods for infrastructural projects.	10	L2	4	1
6. a)	Outline the project management strategies for applying in construction industries.	10	L2	3	1
b)	Explain sustainable construction in reducing negative impacts on environment.	10	L2	4	1
Unit – III					
7. a)	Explain the principles of TERI GRIHA rating for green buildings.	10	L2	5	1
b)	Summarize the concepts in estimation of embodied energy for production of building materials and building envelopes.	10	L2	5	1
8. a)	Explain a case studies on a green building assessed as per LEED ratings.	10	L2	5	1
b)	Explain the concepts in sustainable building design and its economics.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
